

Srushti Sonavane

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SUMMARY

Software Engineer with a track record of building backend and full-stack systems at scale, including public-sector data platforms. At TriFrnd, delivered the Maharashtra State Government teacher-training portal end-to-end for **550,000+** registered teachers across 35 districts; at Barclays, built ETL pipelines moving multi-terabyte financial-crime compliance data from Hadoop to AWS. Core stack: TypeScript, Python, Node.js, PostgreSQL, REST APIs, AWS.

EDUCATION

- Master of Science, Computer Science** | Indiana University Bloomington | GPA: 3.8/4.0
Coursework: Software Engineering, Algorithms, Operating Systems, Databases, Machine Learning
- Aug 2024 – May 2026
- Bachelor of Engg., Computer Science** | Savitribai Phule Pune University
- Sep 2021 – June 2024

EXPERIENCE

- Software Engineer (Graduate Assistant)** | Graduate School Bloomington, Indiana University
- May 2025 – May 2026
- Shipped a live AI chatbot on the Indiana University website serving thousands of student queries per day across **70+** colleges and departments; cut Graduate School office walk-ins by **60%** by answering admissions and general queries instantly
 - The chatbot runs on an A2A multi-agent architecture using Azure LLM and an MCP server for live university data access; a routing agent classifies each query and delegates to department-specific agents, which return the exact department email or phone number for out-of-scope questions; deployed on Azure Kubernetes Service
 - Migrated **13** legacy university applications to shared PHP abstraction layers over XML schemas; verified each migration using automated record-count checks across all affected data
 - Replaced manual form-handling across **70+** university departments with PHP/MySQL pipelines; department coordinators moved from manual data entry to a centralized submission dashboard
- Software Engineer** | TriFrnd (Maharashtra State Government)
- Sep 2023 – Jun 2024
- Owned the teacher-training registration portal end-to-end for the Maharashtra State Government on Next.js, Node.js, and PostgreSQL; the stateless API layer serves concurrent traffic from **550,000+** registered teachers across **35** districts
 - Set up role-based access control across **three permission tiers**; each of the **35** district rollouts was gated by Jest integration tests before going live
 - Wrote a Node.js batch allocation engine that assigns teachers across **960** training batches and **35** districts, replacing a fully manual coordination process with real-time dashboards and audit trails
 - Implemented a two-channel OTP verification flow with eligibility checks that blocked duplicate and invalid submissions across **40,000+** registrations
- Software Engineering Intern** | Barclays
- Jun 2023 – Aug 2023
- Engineered a PySpark ETL pipeline to migrate multi-terabyte financial compliance datasets (FCA, GDPR, and financial crime reporting) from on-premise Hadoop clusters to AWS
 - Packaged reusable PySpark utility modules covering logging, partitioning, and error handling; 2 other data engineering teams adopted them as a standard template

PROJECTS

- AI Course Planning Platform** | *Python, React, FastAPI, LangChain*
- [GitHub](#)
- Created a course recommendation engine using LangChain and the OpenAI API, with a deterministic fallback that ranks by prerequisites, credit load, and student interests when the API is unavailable or rate-limited
 - Coded a React drag-and-drop planner for multi-semester scheduling; prerequisite and credit-limit checks run in the browser so users fix conflicts instantly, with full LLM-generated schedules delivered in under **60 seconds**
- Chicago Food Inspections Platform** | *React, TypeScript, Express, PostgreSQL, PostGIS*
- [GitHub](#)
- Developed the interface and integration layer for a Chicago food-inspection platform, making **250,000+** government records searchable by facility and risk level across Express APIs and PostgreSQL
 - Designed interactive maps and Chart.js dashboards for risk-level analysis, keeping the city-wide map responsive through marker clustering and lazy loading

PUBLICATIONS

- Multilingual Evaluation of Semantic Textual Relatedness (SemEval-2024)*, ACL Anthology 2024
- First Author
- L3Cube-IndicNews: News-based Short Text and Document Classification in Indic Languages*, ACL Anthology 2023
- Co-Author

TECHNICAL SKILLS

- Languages:** Python, TypeScript, JavaScript, SQL, Java, PHP, C++
Backend: Node.js, Express.js, FastAPI, Next.js, RESTful APIs, Microservices, Redis, JWT, RBAC
Frontend: React, HTML, CSS
Databases: PostgreSQL, PostGIS, MySQL, MongoDB, Prisma ORM
Cloud & DevOps: AWS, Azure, Docker, Kubernetes, CI/CD Pipeline, Git, Linux/Unix, PySpark, Hadoop, ETL, Agile, Scrum
AI & LLM: LangChain, OpenAI API, Azure LLM, spaCy, NLTK
Testing: Jest, pytest, Integration Testing